

CHACHOENGSAO AGROMET, Thailand

WMO: 484580

Lat: 13.5672N Lon: 101.4544E Elev: 230 StdP: 14.57 Time Zone: 7.00 (E07) Period: 08-19 WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	61.5	64.1	51.9	57.9	70.9	55.2	65.5	73.6	12.7	76.9	11.4	79.0	0.6	N/A	0.295

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	17.0	96.6	79.2	94.9	78.4	93.5	78.1	81.6	90.3	81.0	89.3	80.6	88.7	4.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.6	155.0	84.3	79.1	152.2	83.8	78.7	150.1	83.4	45.4	90.9	44.9	89.7	44.4	89.0	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.1	8.3	7.0	DB	56.8	98.7	3.3	2.3	54.5	100.3	52.5	101.7	50.7	103.0	48.2	104.6	
			WB	54.7	83.1	3.7	1.3	52.0	84.0	49.8	84.7	47.8	85.5	45.1	86.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.9	78.2	81.0	83.0	84.9	84.7	83.7	82.6	82.7	82.0	81.9	80.6	78.1
	DBStd	3.39	3.74	2.99	2.87	2.80	2.45	1.90	2.08	1.70	1.85	1.77	2.69	3.82
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11660	873	868	1024	1046	1075	1011	1011	1014	961	988	918	871
	CDD65	6185	408	448	559	596	610	561	546	549	511	523	468	406
	CDH74	62982	3787	4561	5990	6680	6821	6152	5719	5627	4853	4899	4256	3637
	CDH80	25500	1549	2050	2784	3170	2896	2393	2067	2029	1649	1776	1671	1466
(14) Wind	WSAvg	1.7	2.0	1.9	2.0	1.6	1.5	1.7	1.8	1.8	1.3	1.2	1.6	2.0
(15) Precipitation	PrecAvg	59.6	0.6	0.8	2.2	3.5	7.0	7.2	8.2	8.9	12.1	7.6	1.4	0.2
	PrecMax	77.1	1.9	2.4	5.4	8.7	11.8	12.8	11.8	15.3	17.8	15.3	5.0	1.2
	PrecMin	46.6	0.0	0.1	0.3	1.2	3.1	3.3	4.4	4.6	6.3	3.3	0.0	0.0
	PrecStd	7.2	0.5	0.6	1.4	2.0	2.4	2.0	2.1	2.4	2.7	2.7	1.2	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.2	95.4	97.7	100.0	97.1	95.7	93.6	92.5	92.9	93.6	93.3	94.0
		MCWB	74.4	76.0	78.6	79.5	79.9	79.2	78.0	78.7	80.1	79.3	77.0	75.7
	2%	DB	91.4	93.7	95.4	97.3	95.2	92.9	91.6	91.0	91.0	91.4	91.7	91.9
		MCWB	73.7	75.5	77.8	79.7	79.7	79.3	78.4	78.8	79.5	79.1	77.3	75.5
	5%	DB	89.9	92.0	93.5	95.0	93.5	91.2	90.1	89.8	89.5	89.9	90.3	89.9
		MCWB	73.1	75.1	77.6	79.5	79.7	79.2	78.4	78.6	79.2	78.9	77.3	74.8
	10%	DB	88.1	90.2	91.7	92.7	91.4	89.7	88.5	88.4	87.9	88.3	88.5	87.7
		MCWB	73.0	74.9	77.6	79.4	79.8	79.1	78.5	78.5	79.0	78.7	77.0	74.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.8	79.3	80.6	81.9	82.5	81.9	80.9	81.0	81.4	81.3	80.7	79.2
		MCDB	87.4	89.8	92.0	93.1	91.5	88.7	87.5	87.7	88.8	89.0	88.8	87.3
	2%	WB	77.5	78.4	79.8	81.3	81.6	81.0	80.1	80.2	80.7	80.3	79.7	78.0
		MCDB	85.4	87.9	90.7	91.9	90.2	88.2	86.8	86.9	87.9	87.8	87.4	86.3
	5%	WB	76.6	77.8	79.1	80.7	80.9	80.5	79.6	79.7	80.0	79.8	79.0	77.0
		MCDB	84.0	86.9	89.6	90.8	89.0	87.7	86.1	86.2	86.8	87.0	86.3	85.2
	10%	WB	75.6	77.1	78.5	80.0	80.3	79.9	79.1	79.1	79.3	79.2	78.2	75.9
		MCDB	82.4	85.9	88.2	89.3	87.8	86.5	85.4	85.4	85.7	86.0	84.9	84.1
(35) Mean Daily Temperature Range	5% DB	MDBR	20.4	19.9	18.1	17.0	14.0	12.7	11.9	12.4	12.3	13.5	16.0	19.2
		MCDBR	21.9	21.7	20.7	19.5	16.7	14.8	14.0	14.0	14.6	15.8	18.1	21.0
		MCWBR	6.7	6.0	5.6	4.5	4.0	3.8	3.6	3.9	4.6	5.2	5.5	6.5
	5% WB	MCDBR	17.6	18.8	18.4	17.6	14.8	13.1	12.2	12.9	13.6	14.0	15.5	18.0
		MCWBR	6.2	5.5	5.5	4.9	4.6	4.4	4.3	4.5	4.9	5.1	5.4	6.2
(40) Clear-Sky Solar Irradiance	taub		0.534	0.612	0.632	0.572	0.496	0.474	0.469	0.471	0.488	0.546	0.490	0.479
	taud		1.962	1.795	1.788	1.971	2.233	2.293	2.309	2.289	2.230	2.032	2.172	2.154
	Ebn at Noon		243	229	229	242	257	260	262	265	262	244	254	255
	Edh at Noon		57	70	72	60	45	42	42	43	46	55	46	46
(44) All-Sky Solar Radiation	RadAvg		1567	1627	1658	1749	1597	1540	1469	1480	1451	1496	1548	1544
(45)	RadStd		100	80	91	94	89	63	74	90	73	88	94	64

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (31 neighbors)	N/A	N/A	N/A	N/A	+0.43	+0.45	N/A	N/A	+133	+133

Nomenclature: See separate page